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PTH STA

PlasticO:
An ontology for plastic and plastic pollution

Giorgio A. Ubbiali and Ludger Jansen

PTH Brixen College

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Plastics and Plastic Pollution

- Plastics are largely used and beneficial in disparate fields, including constructions, agriculture, and daily use.

- Several pollution concerns.
  - Alteration of soil biogeochemical element cycles.
  - Greenhouse gas emissions.
  - Water contamination and trophic transfer of microplastics.
  - Microplastic infiltration of organisms' bodies.

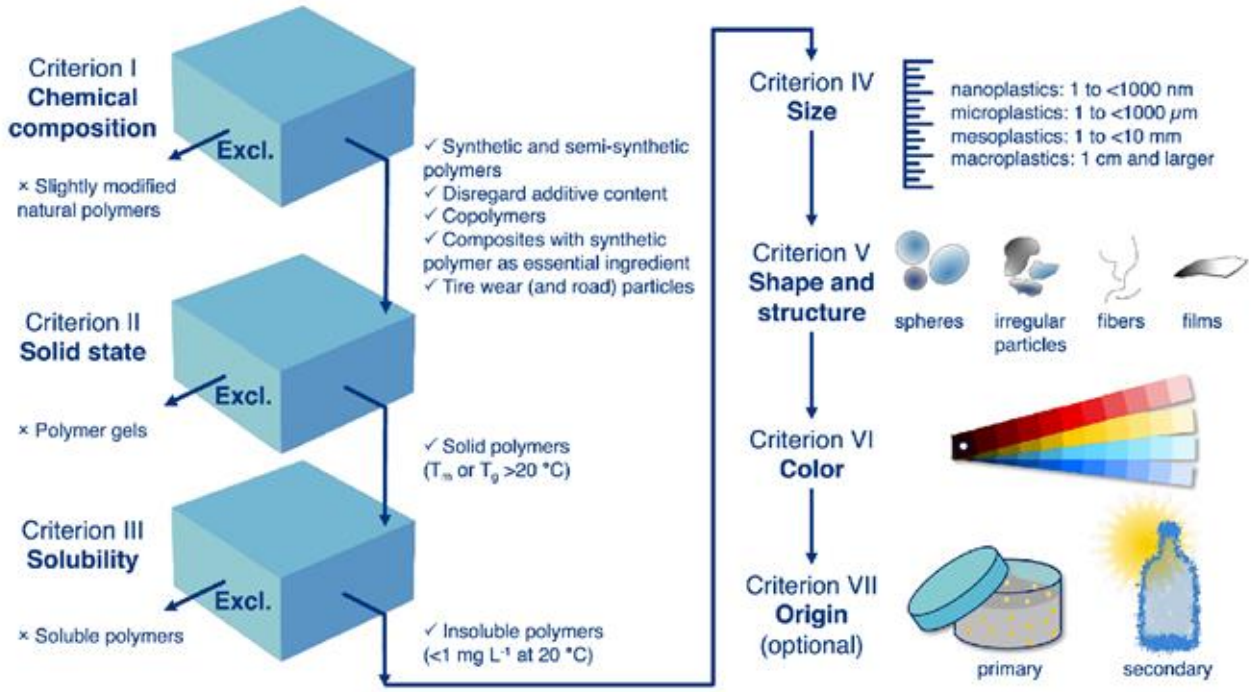
  Mutual interconnections with climate change.

(Eerkes-Medrano & Thompson, 2018 ; FAO, 2021; Filipe et al., 2023; Hartmann et al., 2019; Kelly et al., 2025; Lakhiam et al., 2024; Masterton et al., 2023; Ojukwu et al., 2024; Rochman et al., 2019 Yu et al., 2022; Yuan et al., 2022)



Plastics and Climate Change Semantics

- Cross-domain studies.
  - Vague terminologies.
  - Disparate standards of reference.
  - Various systems of classification.



An account of plastic debris. From Hartman et al. (2019).

*An attractive opportunity: **Ontologies**.*

But to date, no ontology addresses plastic pollution and its interrelations with climate change.

The ClimOO Ontology Suite

- A reference ontology suite for representing plastics, (micro-)plastic pollution, climate change, and their interrelations.
- Use case: agrifood systems; South Tyrolean (Italy) apple production.

- Ontology Repo: https://w3id.org/climoo/repo
- ClimOO Project: https://pthsta.it/climoo
- English + Web Ontology Language (OWL).



Source:https://pthsta.it/climoo

The ClimOO Suite Architecture

- A modular structure:
  - Plastics Ontology (PlasticO),
  - ClimOO-SDC2,
  - Plastics Interconnecting with Climate Change Ontology (PICCO),
  - PICCOmela.
- Extending Basic Formal Ontology (BFO) and Sustainability Core Ontology (SCO).

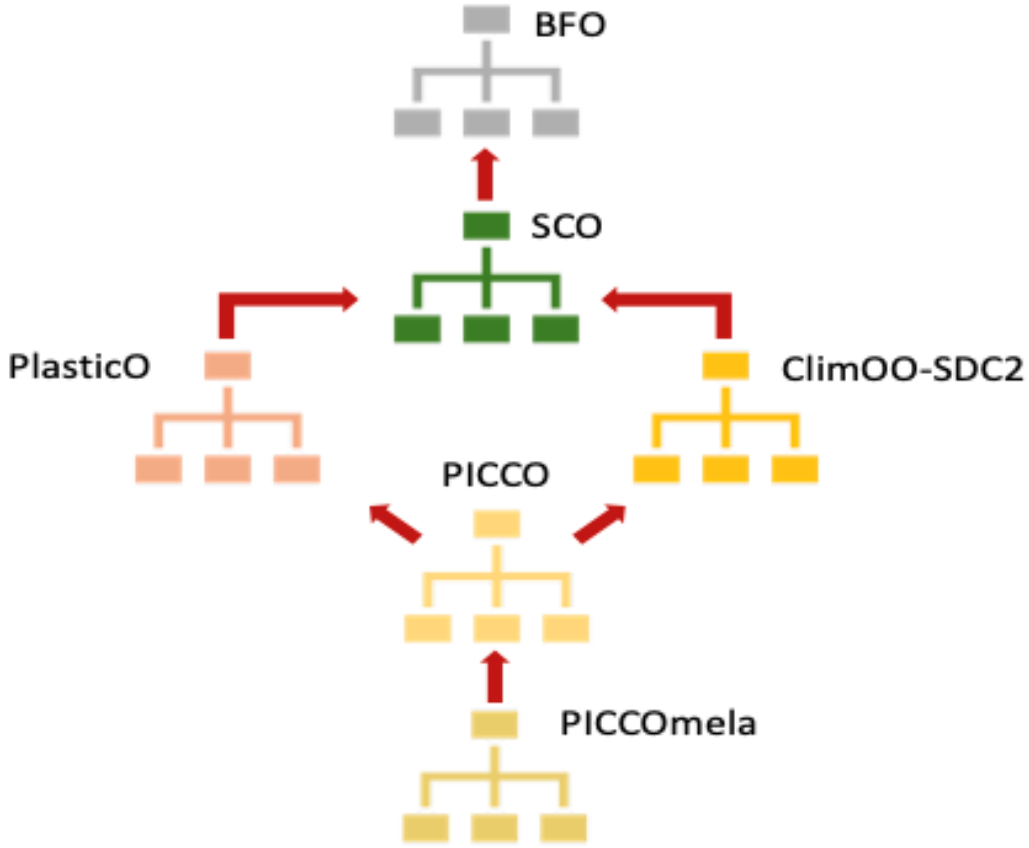


Fig. The ClimOO Ontology suite structure. From Ubbiali and Jansen (2026b).

The ClimOO Methodology

- Sustainable Meat Systems Ontology (SuMSO) methodology (Ubbiali et al., 2026).
  - Based on existing standards and methodologies such as METHONTOLOGY and the OTK methodology.

- Assessment of the relevant literature and major international standards.
- Stakeholder engagement.
- Reasoner validation and Competency Questions (CQs).
- Imports from existing resources, including EnvO, ChEBI, and PMDco.
- OBO Foundry principles.

(Fernández-López et al., 1997; Staab et al., 2001; Ubbiali et al., 2026)



The PlasticO Module

- Domain of reference: plastics and plastic life cycles, plastic debris, and plastic pollution processes.
- Transdisciplinary approach.

- Extending SCO and BFO and importing from 14 ontologies, mainly from the OBO Foundry and BFO environments.
- 463 classes, 273 object properties, and 8 individuals.
- https://www.w3id.org/climoo/climoo-plastico

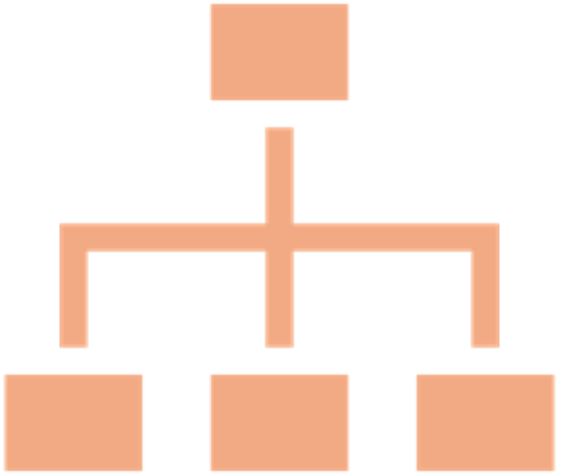
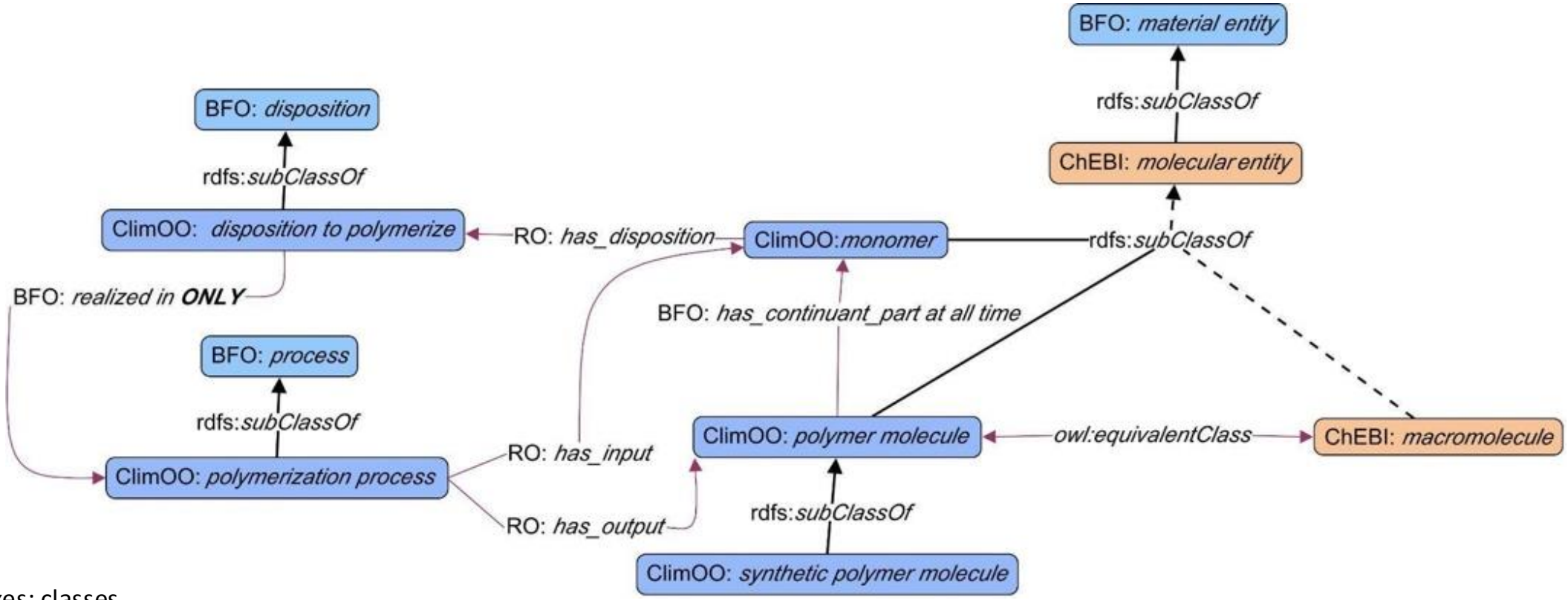


Fig. The PlasticO module.

Polymers Molecules



Boxes: classes
Bold arrows: indirect is_a
Dashed arrows: indirect is_a
Red arrows: other relations

Fig. Polymers and monomers From Ubbiali and Jansen (2026a).

Plastics and Pieces of Plastics

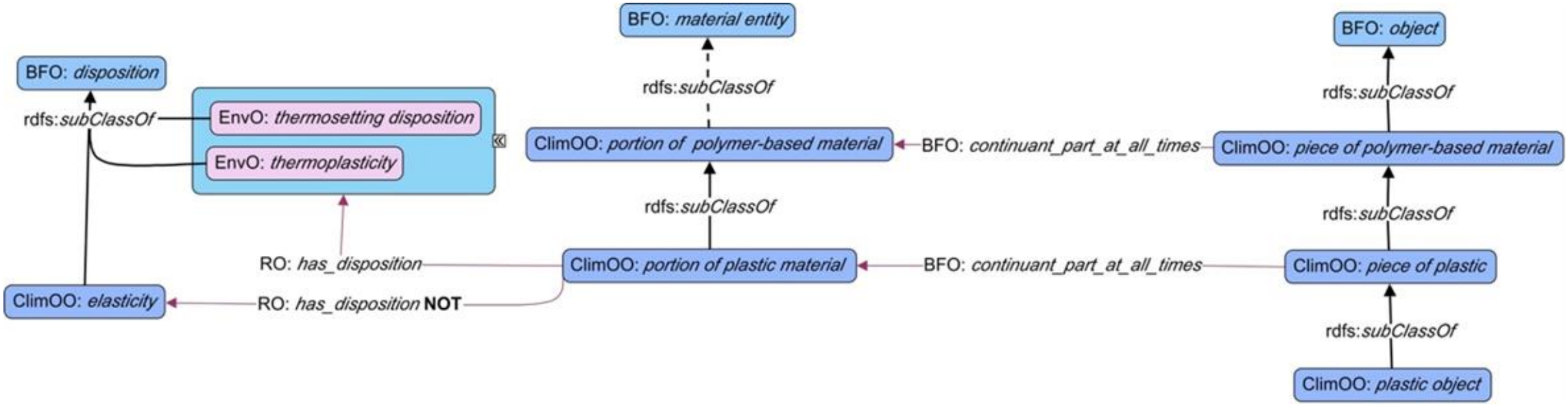


Fig. Portions and pieces of plastic. From Ubbiali and Jansen (2026a).

Plastic Life Cycle

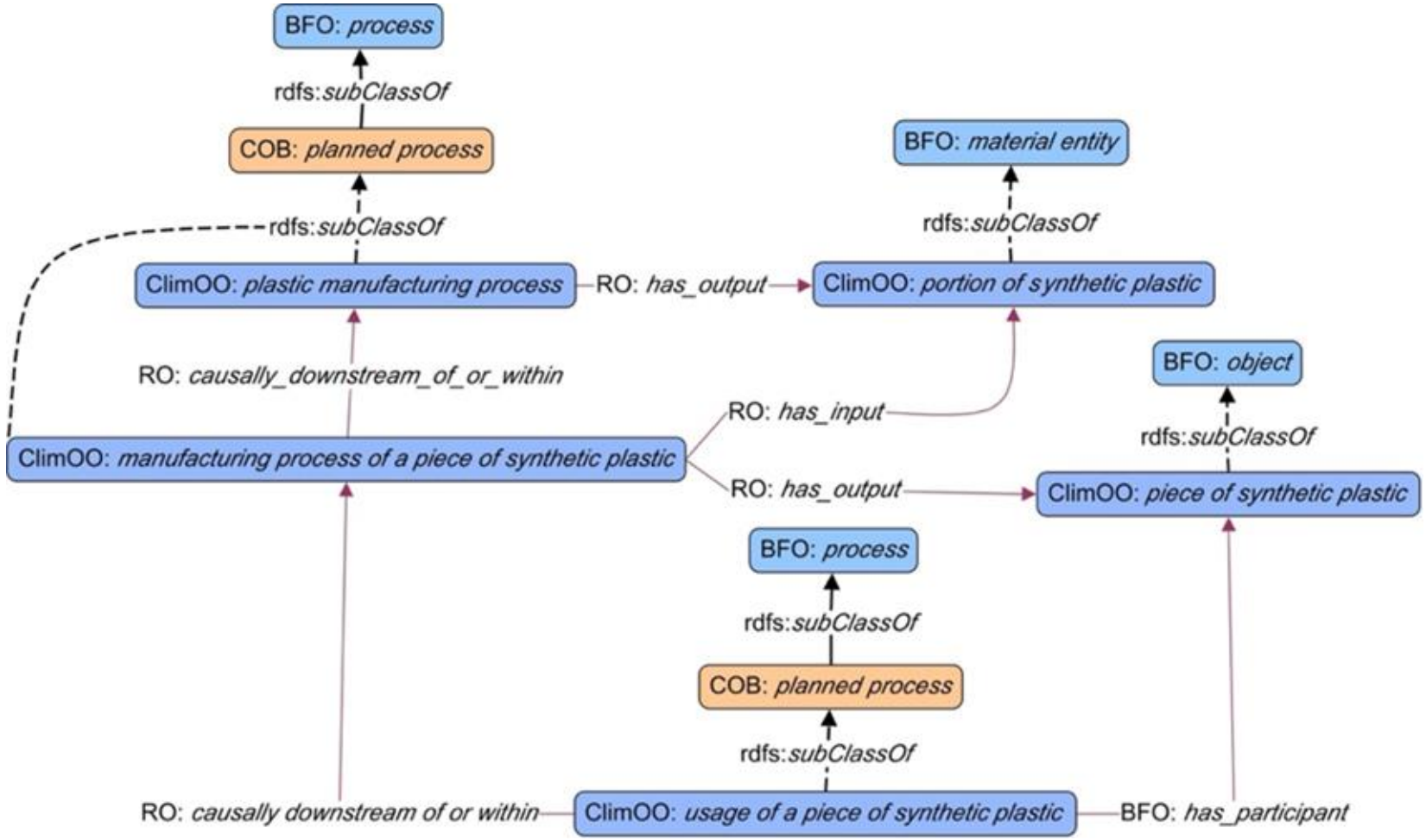


Fig. Plastic life cycle. From Ubbiali and Jansen (2026b).

Plastic Debris

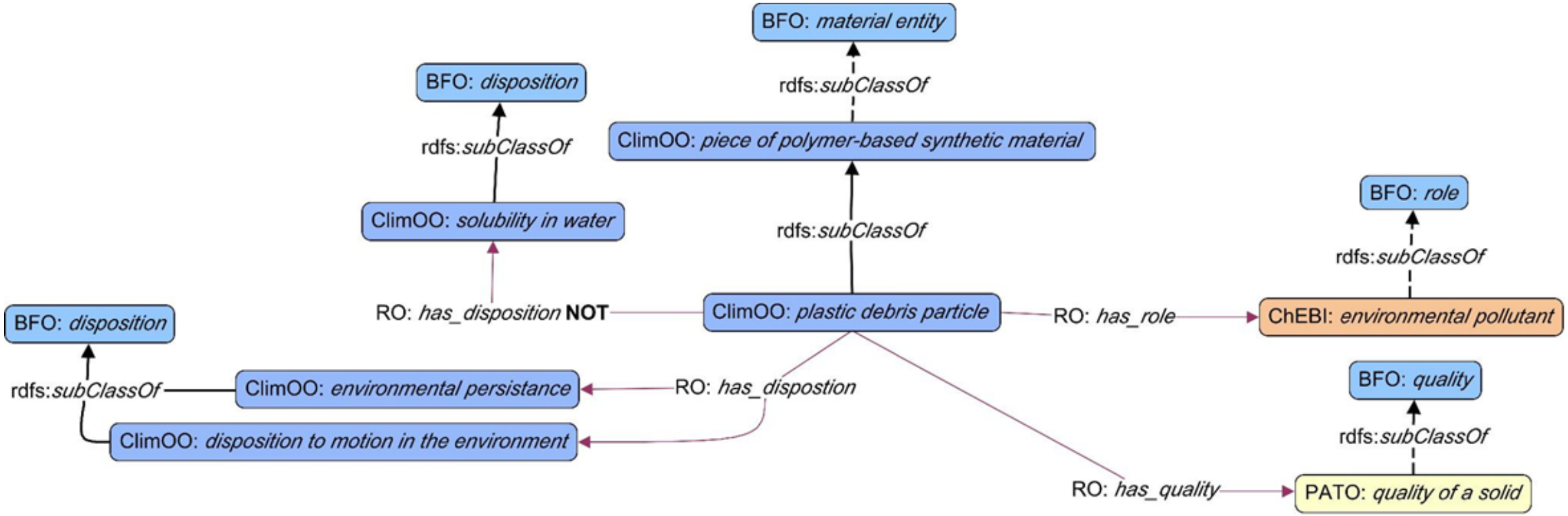


Fig. Plastic debris particles. From Ubbiali and Jansen (2026a).

Plastic Pollution

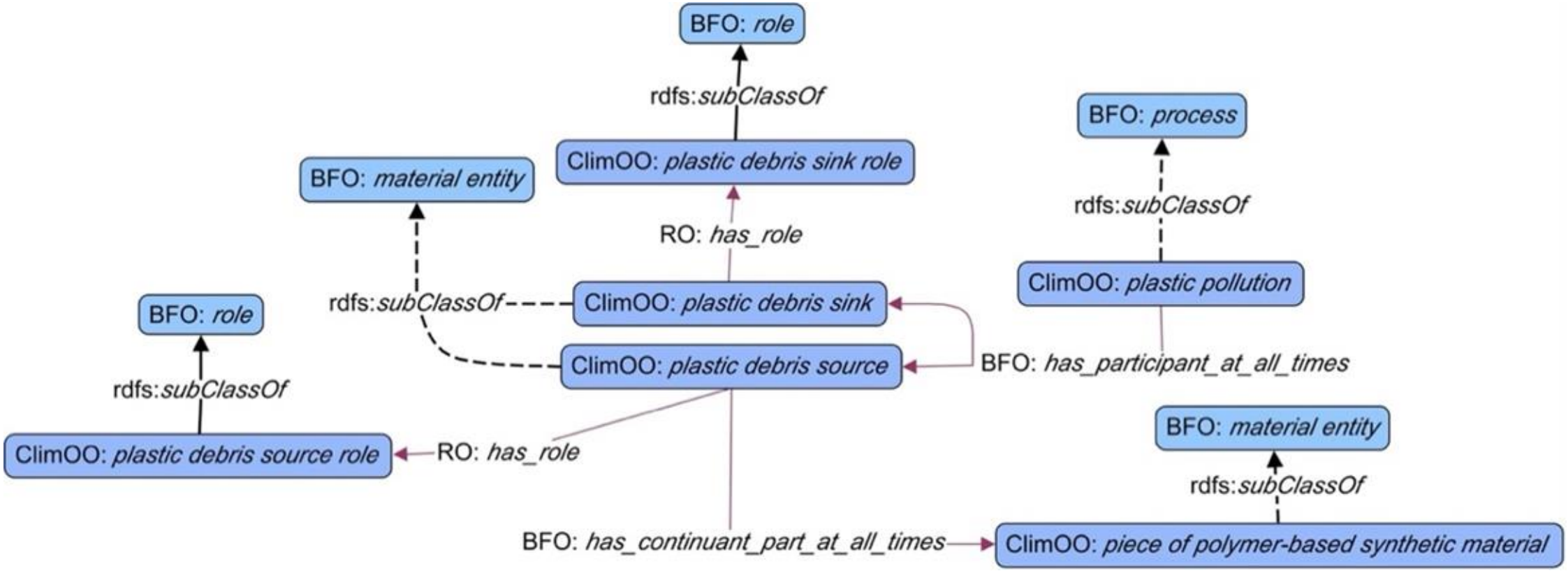


Fig. Plastic pollution. From Ubbiali and Jansen (2026b).

Conclusions and Ongoing Works

- The design of an ontology suite for plastic pollution and climate change: ClimOO.
- Ongoing works:
  - Construction of the PICCO and PICCOmela modules.
  - SPARQL queries for CQs testing.
  - Stakeholder approval.
  - Use case datasets for ontology implementation.

COSMOS: A Community of Common Ontologies, Semantics, and MOdels for Sustainability.



References and Slide Deck



https://github.com/ClimOO/ClimOO-Ontology/tree/main/ClimOO%20material/materials%20from%20ClimOO%20presentations/5th%20Ontologies4Chem%20Workshop



Thank You!!

Giorgio.Ubbiali@pthsta.it